

Control Unit Design and Working

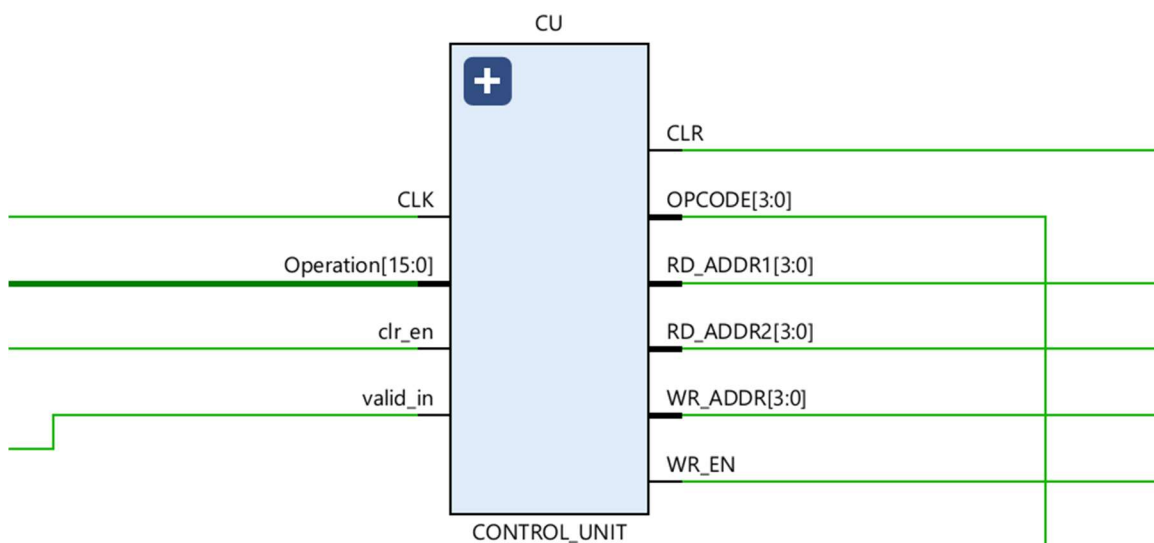
Introduction

The Control Unit is responsible for coordinating the operation of the pipelined ALU system. It receives instructions from the external interface, decodes instruction fields, generates control signals, and ensures that data, addresses, and operation codes move through the pipeline in a synchronized manner.

Purpose of the Control Unit

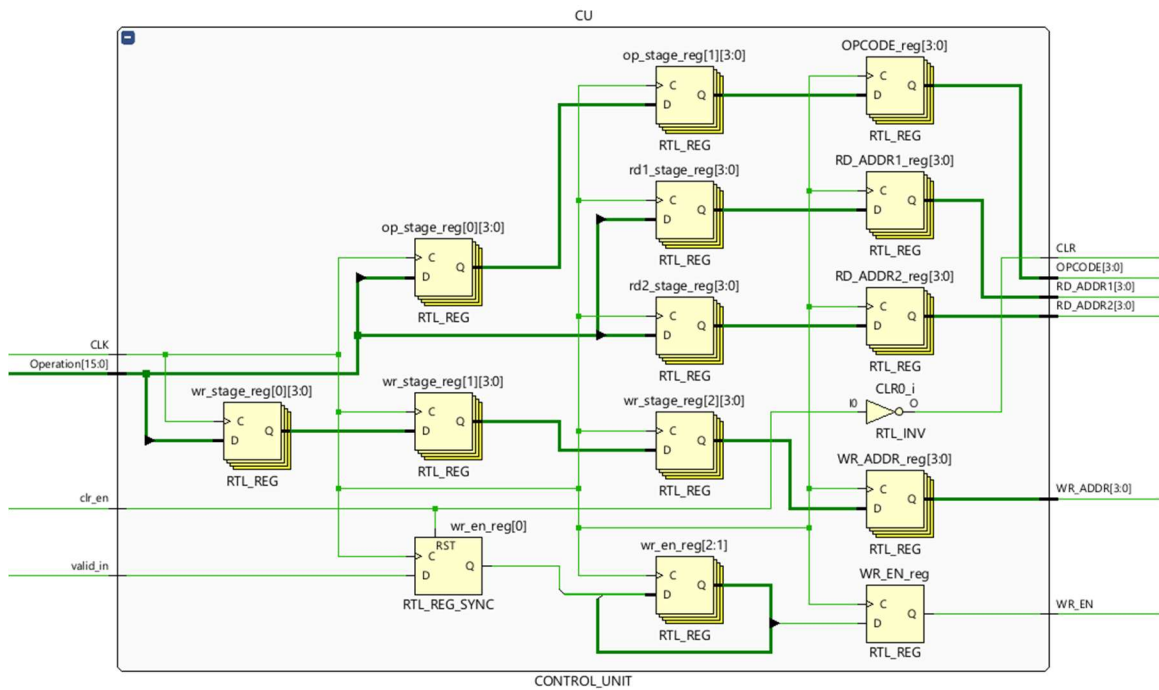
The Control Unit acts as the central coordinator of the design. Its primary responsibilities include instruction decoding, generation of register file read and write addresses, opcode extraction, pipeline synchronization, write-enable generation, and reset control.

Instruction Format



The Control Unit receives a 16-bit instruction. The instruction contains an opcode field and register addressing fields. After decoding, the opcode is forwarded toward the ALU while source and destination addresses are routed toward the register file and writeback logic.

Circuit Overview



Pipeline Coordination

The design follows a pipelined architecture. The Control Unit does not execute arithmetic operations itself. Instead, it ensures that the correct opcode, source register addresses, destination register address, and control signals remain aligned as the instruction moves through the pipeline stages.

Pipeline Stages

Stage 1: Instruction Decode. The instruction is decoded and latched.

Stage 2: Register Read. Source operands are fetched from the register file.

Stage 3: Execute. The ALU performs the selected operation and generates flags.

Stage 4: Writeback. The computed result is written to the destination register.

Address Propagation

The destination register address must travel through dedicated pipeline registers so that it reaches the register file write port at the same time as the corresponding ALU result. This prevents writing valid data into an incorrect destination location.

Opcode Handling

The opcode extracted during instruction decode is forwarded towards the ALU. The ALU uses the opcode to select one of the supported operations such as ADD, SUB, shift operations, logical operations, inversion operations, buffer operations, and comparison operations.

Control Signals

Typical control signals generated or managed by the Control Unit include VALID_IN, CLEAR, WRITE_ENABLE, source register addresses, destination register address, and opcode outputs.

Reset Behavior

The Control Unit supports reset functionality to place the design into a known state. During reset, pipeline contents are cleared.

Writeback Synchronization

A critical responsibility of the Control Unit is ensuring that write enable signals are synchronized with the correct destination address and ALU result. All writeback-related signals must experience identical pipeline latency.

Verification Strategy

The Control Unit is verified using directed SystemVerilog testbenches. Verification confirmed correct instruction decoding, address propagation, opcode propagation, reset behavior, write-enable timing, and pipeline synchronization.

Conclusion

The Control Unit forms the supervisory component of the pipelined ALU architecture. Its primary objective is to maintain correct timing relationships between instructions, operands, addresses, results, and control signals throughout the pipeline.